HKUST – Department of Computer Science and Engineering
COMP 2711: Discrete Math Tools for Computer Science

Fall 2025 Midterm Examination

Date: 29 Oct 2025 Time: 9:00-11:00

Name: _____	Student ID: _____
Email: _____	Lecture: _____

Instructions

- This is a closed book exam. It consists of **10** questions. You are allowed to bring an A4 single-sided, handwritten, cheat sheet.
- Please write your name, student ID, email, and lecture section in the space provided at the top of this page.
- Please sign the honor code statement on page 2.
- Answer all questions within the space provided on the examination paper. You may also use the back of the pages. Each question is on a separate page. This is for clarity and is not meant to imply that each question requires a full-page answer. Some can be answered using only a few lines.
- Solutions can be written in terms of binomial coefficients, factorials, and the $C(n,k)$, $P(n,k)$ notation. For example, you can write $\binom{5}{3}+\binom{4}{2}$ instead of 16. Avoid using nonstandard notations such as ${}_nP_k$ or ${}_nC_k$.

Problems	1	2	3	4	5	6	7	8	9	10	Total
Points	10	13	12	16	10	9	7	6	7	10	100
Score											

As part of HKUST's introduction of an honor code, the HKUST Senate has recommended that all students be asked to sign a brief declaration printed on examination answer books that their answers are their own work, and that they are aware of the regulations relating to academic integrity. Following this, please read and sign the declaration below.

I declare that the answers submitted for this examination are my own work.

I understand that sanctions will be imposed, if I am found to have violated the University regulations governing academic integrity.

Student's Name: _____

Student's Signature: _____

Problem 1: Propositional logic, Inference and proofs [10 points]

We define NAND operator ($\overline{\wedge}$) as $p \mathbin{\overline{\wedge}} q = \neg\,(p \wedge q)$.

(a) **[3 points]** Use propositional laws to prove that

$$p \mathbin{\overline{\wedge}} (q \rightarrow r) \equiv (p \rightarrow q) \wedge (p \rightarrow \neg r)$$

(you are **NOT** allowed to use truth table)

(b) **[2 points]** Argue whether the following two expressions are equivalent or not. If they are, prove it without using a truth table. If they are not, provide a counterexample.

Expression A: $p \mathbin{\overline{\wedge}} (p \mathbin{\overline{\wedge}} q)$

Expression B: $(p \mathbin{\overline{\wedge}} p) \mathbin{\overline{\wedge}} q$

(c) **[5 points]** Let p, q, r, t denote four propositions. You are given the following premises:

1.$p \to (r \to q)$
2. $\neg q$
3. $(q \vee p) \wedge t$

Using rules of inference for propositional logic, show that the truth of the above premises implies that the conclusion "$\neg r \wedge t$" is true.

Answer: (a) Start with the left-hand side:

$$p \overline{\wedge} (q \to r) = \neg(p \wedge (q \to r)).$$

Since $q \to r \equiv \neg q \vee r$:

$$p \wedge (q \to r) = p \wedge (\neg q \vee r).$$

Apply De Morgan's law to the NAND:

$$\neg(p \wedge (\neg q \vee r)) = \neg p \vee \neg(\neg q \vee r) = \neg p \vee (q \wedge \neg r).$$

Now consider the right-hand side:

$$(p \to q) \wedge (p \to \neg r) = (\neg p \vee q) \wedge (\neg p \vee \neg r).$$

Apply the distributive law:

$$(\neg p \vee q) \wedge (\neg p \vee \neg r) = \neg p \vee (q \wedge \neg r).$$

Both sides are equal:

$$\neg p \vee (q \wedge \neg r) \equiv \neg p \vee (q \wedge \neg r).$$

Thus, $p \overline{\wedge} (q \to r) \equiv (p \to q) \wedge (p \to \neg r)$.

(b) Consider a counterexample: let $p =$ true, $q =$ false.

- Expression A: False. - Expression B: True.

Since the expressions differ, they are not equivalent.

(c)

1. $p \to (r \to q)$ premise
2. $\neg q$ premise
3. $(q \vee p) \wedge t$ premise
4. t from 3
5. $q \vee p$ from 3
6. p from 5, 2
7. $r \to q$ from 6, 1
8. $\neg r \vee q$ from 7
9. $\neg r$ from 8, 2
10. $t \wedge \neg r$ from 9,4

Problem 2: Predicate logic [13 points]

(a) **[4 points, 2 pts each]** Consider the set of Natural numbers $\mathbb{N}$ to be the domain. Express the following sentences using only $\forall, \exists, \wedge, \vee, \rightarrow, \leftrightarrow, \neg, =, \neq, <, >, +, \times$.

(i) x is a multiple of 5.

(ii) x is a perfect square number. (a perfect square number x can be written as $x = y^2$ for an integer y)

(b) **[4 points]** Express the negations of each of these statements so that all negation symbols immediately precede predicates.

$$\forall x \ (p(x) \quad \vee \quad \exists y(p(y) \rightarrow \forall z \ q(x, z)))$$

(c) **[5 points]** Consider the following predicates.

$S(x)$ be "x is a satisfactory"

$C(x)$ be "x is a clear explanation"

$E(x)$ be "x is an excuse"

Assume the following premises:

Premise 1: All clear explanations are satisfactory

Premise 2: Some excuses are unsatisfactory

First, write each premise as a logical form using quantifiers and the above predicates and then write a formal proof for the following statement:

Some excuses are not clear explanations.

Answer: (a) (i) $\exists y \;\; s.t. \;\; 5 \cdot y = x.$
(ii) $\exists y \;\; s.t. \;\; y \cdot y = x.$

(b)

$$\neg\forall x \; (p(x) \vee \exists y \; (p(y) \rightarrow \forall z \; q(x,z))) = \exists x \; \neg(p(x) \vee \exists y \; (p(y) \rightarrow \forall z \; q(x,z))).$$

Apply De Morgan's law to the disjunction:

$$\neg(p(x) \quad \vee \exists y \; (p(y) \rightarrow \forall z \; q(x,z))) = \neg p(x) \wedge \neg\exists y \; (p(y) \rightarrow \forall z \; q(x,z)).$$

Negate the existential quantifier:

$$\neg\exists y \; (p(y) \rightarrow \forall z \; q(x,z)) = \forall y \; \neg(p(y) \rightarrow \forall z \; q(x,z)).$$

Rewrite the implication $p(y) \rightarrow \forall z \; q(x,z) = \neg p(y) \vee \forall z \; q(x,z)$, so:

$$\neg(p(y) \rightarrow \forall z \; q(x,z)) = \neg(\neg p(y) \vee \forall z \; q(x,z)) = p(y) \wedge \neg\forall z \; q(x,z).$$

Negate the universal quantifier:

$$\neg\forall z \; q(x,z) = \exists z \; \neg q(x,z).$$

Thus:

$$\neg(p(y) \rightarrow \forall z \; q(x,z)) = p(y) \wedge \exists z \; \neg q(x,z).$$

So:

$$\neg\exists y\ (p(y) \to \forall z\ q(x,z)) = \forall y\ (p(y) \wedge \exists z\ \neg q(x,z)).$$

Combine:

$$\exists x\ \left(\neg p(x) \wedge \forall y\ (p(y) \wedge \exists z\ \neg q(x,z))\right).$$

This expression has all negation symbols immediately preceding predicates ($\neg p(x)$ and $\neg q(x,z)$).

$$\boxed{\exists x\ \left(\neg p(x) \wedge \forall y\ \ (p(y) \wedge \exists z\ \neg q(x,z))\right)}$$

(c) The premises turn into:

$$\forall x\,(C(x) \to S(x))$$

$$\exists x\,E(x) \wedge \neg S(x).$$

While conclusion is $\exists x\,E(x) \wedge \neg C(x)$.

The proof goes as follows:

1. $\exists x\,E(x) \wedge \neg S(x)$ (premise)
2. $E(x_0) \wedge \neg S(x_0)$ (for some x_0, by existential specification of 1))
3. $\forall x\,(C(x) \to S(x))$ (premise)
4. $C(x_0) \to S(x_0)$ (by universal specification of 3))
5. $\neg S(x_0)$ (by simplification of 2))
6. $\neg C(x_0)$ (Modus Tollens of 4) and 5))
7. $E(x_0)$ (by simplification of 2))
8. $E(x_0) \wedge \neg C(x_0)$ (conjunction of 6) and 7))
9. $\exists x\,E(x) \wedge \neg C(x)$ (existential generalization of 8))

Problem 3: Sets and Functions [12 points]

For a set A, the complement of A denoted by $\overline{A}$ is given by $\{x \mid x \notin A\}$

(a) **[2 points]** Assume $A \subseteq C$ and $B \subseteq C$. Use direct proof to prove that $A \cap B \subseteq C$.

(b) **[3 points]** Assume $A \subseteq C$ and $B \subseteq C$. Use proof by contrapositive to show that $A \cup B \subseteq C$.

(c) **[7 points, 1 pt each]** Determine whether the following statements are True or False. **No argument is needed**.

(i) for any set $S \subset \mathbb{N}$, $P(S)$ is uncountable infinite.

(ii) The set of functions from $\mathbb{N} \to \mathbb{N}$ is uncountable infinite.

(iii) Assume A and B are both uncountable sets such that $A \subset B$. $B - A$ is either countable infinite or finite.

(iv) Set S is countable infinite if there exists a bijection between S and the set of integers $\mathbb{Z}$.

(v) The following set is countable infinite $\{0, 00, 000, 0000, \cdots\}$.

(vi) The set of binary strings of infinite length is uncountable.

(vii) The set of binary strings of infinite length such that it has exactly three "1"'s is countable infinite.

Answer: (a) Let $x \in A \cap B$. By definition of intersection, $x \in A$ and $x \in B$. Since $A \subseteq C$, it follows that $x \in C$. Similarly, since $B \subseteq C$, $x \in C$ also holds. Thus, $x \in C$. Therefore, $A \cap B \subseteq C$.

(b) We prove the contrapositive: Assume $A \cup B \not\subseteq C$, which means there exists an x such that $x \in A \cup B$ and $x \notin C$. which means $x \in A \vee x \in B$ and $x \notin C$. This means $(A \not\subseteq C)$ OR $(B \not\subseteq C)$ Which is the negation of $(A \subseteq C\ )$ AND $(B \subseteq C)$

(c) (i) F

(ii) T

(iii) F

(iv) T

(v) T

(vi) T

(vii) T

Problem 4: Counting [16 points]

Answer the following questions. You can leave your answer in a combinatorial form without calculating the final answer. For example something like $5! \times 2^3 \times P(5,3) \times 2^7 \times \binom{20}{2}$ is totally fine. **No explanations is needed.** You are **NOT** allowed to use $\sum$.

(a) **[8 points, 2 pt each]** Assume having 6 distinct teachers and 5 distinct students.

(i) In how many ways they can stand in a line?

(ii) In how many ways, all teachers are adjacent to each other?

(iii) In how many ways, all teachers are adjacent and also all students are adjacent to each other?

(iv) In how many ways, no two teacher stand next to each other.

(b) **[2 point]** In how many ways you can go from A to D and come back to A such that you can only use a way once.



(c) **[4 point, 2 pts each]**
Imagine there are two sets $A = \{a_1, a_2, \cdots, a_7\}$ and $B = \{b_1, b_2, \cdots, b_{10}\}$. Answer the following questions:

(i) In how many ways we can have 7 unordered pairs (e.g., (a_1, b_1) is the same as (b_1, a_1)) such that each pair contains an element from A and an element from B while the elements can not be reused?

(ii) In how many ways we can have 5 unordered pairs (e.g., (a_1, b_1) is the same as (b_1, a_1)) such that each pair contains an element from A and an element from B while the elements can not be reused?

(d) **[2 point]** We call a 5-digit number *palindrome* if it is of form $\overline{abcba}$. For example 11111 and 12421 are both palindromes. However, 21013 is not. How many 5-digit palindrome numbers exist?
Note that the leftmost digit can not be 0.

Answer: (a) (i) $11!$
(ii) $6! \times 6!$
(iii) $2! \times 5! \times 6!$
(iv) $6! \times 5!$

(b) $3 \times 4 \times 5 \times 4 \times 3 \times 2$

(c) (i) $P(10, 7)$
(ii) $\binom{7}{5} \times \binom{10}{5} \times 5!$

(d) $9 \times 10 \times 10$

Problem 5: Combinatorial Proof [10 points]

Use combinatorial reasoning to prove the identity.

$$\binom{n}{k} - \binom{n-3}{k} = \binom{n-1}{k-1} + \binom{n-2}{k-1} + \binom{n-3}{k-1}.$$

Answer: Let S be a set of n elements, and let a, b, c be distinct elements of S. The number of k-subsets of S is $\binom{n}{k}$, and the number of k-subsets of $S - \{a, b, c\}$ is $\binom{n-3}{k}$. Then the LHS is the number of k-subsets of S that contain at least one of the elements of $\{a, b, c\}$. Such k-subsets can be divided into 3 types: (1) the k-subsets that contain the element a; (2) the k-subsets that do not contain a but contain b; and (3) the k-subsets that do not contain a, b but contain c. The numbers k-subsets of type (1), type (2), type (3) are

$$\binom{n-1}{k-1}, \quad \binom{n-2}{k-1}, \quad \binom{n-3}{k-1}$$

respectively. The sum of these numbers is exactly the RHS.

Problem 6: Inclusion-Exclusion [9 points]

Use inclusion-exclusion principle to solve the following problem. No other methods are allowed. You do not need to calculate the final value.

In how many 6-digit numbers do all three digits 1, 2, and 3 appear at least once? Note that the leftmost digit can not be equal to 0.

Answer: Assume S to be the set of all 6-digit numbers.

$$|S| = 9 \times 10^5.$$

Let A_1 be the set of 6-digit numbers missing digit 1, A_2 be the set of 6-digit numbers missing digit 2 and A_3 be the set of 6-digit numbers missing digit 3 . We need the number of 6-digit numbers where digits 1, 2 and 3 appear, which is the total minus the numbers missing at least one of 1 or 2 or 3:

$$S - (A_1 \cup A_2 \cup A_3).$$

By inclusion-exclusion:

$$|A_1 \cup A_2 \cup A_3| = |A_1| + |A_2| + |A_3| - |A_1 \cap A_2| - |A_1 \cap A_3| - |A_2 \cap A_3| + |A_1 \cap A_2 \cap A_3|.$$

- $|A_1|$: Numbers with no 1. First digit has 8 choices (2 to 9), other digits have 9 choices (0, 2 to 9):

  $$|A_1| = 8 \times 9^5.$$

  Similarly, $|A_1| = |A_2| = |A_3| = 8 \times 9^5$.
- $|A_1 \cap A_2|$: Numbers with no 1 and no 2. First digit has 7 choices (3 to 9), other digits have 8 choices (0, 3 to 9):

  $$|A_1 \cap A_2| = 7 \times 8^5.$$

  Similarly, $|A_2 \cap A_3| = |A_1 \cap A_3| = |A_1 \cap A_2| = 7 \times 8^5$.
- $|A_1 \cap A_2 \cap A_3|$: Numbers with no 1 and no 2 and no 3. First digit has 6 choices (4 to 9), other digits have 7 choices (0, 4 to 9):

  $$|A_1 \cap A_2 \cap A_3| = 6 \times 7^5.$$

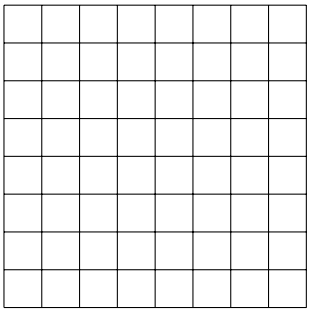
Now compute:

$$|A_1 \cup A_2 \cup A_3| = 3 \times (8 \times 9^5) - 3 \times (7 \times 8^5) + 6 \times 7^5$$

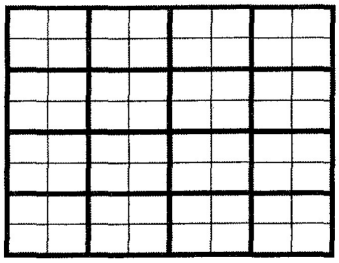
So the answer is $|S| - |A_1 \cup A_2 \cup A_3| = 9 \times 10^5 - 3 \times (8 \times 9^5) + 3 \times (7 \times 8^5) - 6 \times 7^5$

Problem 7: Pigeonhole Principle [7 points]

Consider an 8×8 grid of 1×1 cells. If 17 cells are colored red at random, prove that there exist at least two red cells that share a common corner.



Answer: We split the 8×8 grid into 16 2×2 grids as follows:



Note that any two cells from a 2×2 Grid contains a shared common corner. If we color 17 1×1 cells, it means we color at least two cells from one 2×2 grid which have a shared common corner.

Problem 8: Probability [6 points] A spider has $\frac{1}{2}$ probability of successfully catching every fly it encounters during hunting. Let k be the number of flies that has been caught so far by the spider, after it has encountered 5 flies. Answer the following questions:

(a) **[1 points]** Calculate $P(k = 0)$

(b) **[1 points]** Calculate $P(k = 1)$

(c) **[1 points]** Calculate $P(k = 2)$

(d) **[1 points]** If the spider needs to eat 3 flies per meal, what is the probability that it is still hungry after encountering 5 flies? (The spider stops hunting after eating 3 flies)

(e) **[2 points]** Assume the spider encounters a sixth fly. What is the probability that the 6th fly it encounters survives?



Spider: Good luck! I suggest you also to have some snacks or water to get some energy:)

Answer: Lets calculate the probability that the spider is still hungry . Let k denote the number of flies that it has eaten so far.

$P(k=0) = \frac{1}{2^5}$. $P(k=1) = \binom{5}{1} \cdot \frac{1}{2^5}$ $P(k=2) = \binom{5}{2} \cdot \frac{1}{2^5}$

So $P(Hungry) = \frac{1+5+10}{32} = \frac{1}{2}$ Which means $P(full) = 1 - P(Hungry) = \frac{1}{2}$.

So we can conclude that $P(Sixth_Fly_Survive) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot 1 = \frac{3}{4}$

has not eaten anyone of them.

Problem 9: Conditional Probability [7 points]

Two coins look similar to each other. One of them is a fair coin: $P(heads) = \frac{1}{2}$. The other coin is a biased coin: $P(heads) = \frac{2}{3}$. We pick one of the coins at random and toss it once, resulting in heads. Then we pick the other coin and toss it three times, resulting in two heads.

Let events D and B be defined as follows:

D: first coin head in 1 toss and second coin 2 heads in 3 tosses. (Basically it denotes the observation that we have seen)

B: first coin is biased

(a) **[3 points]** Calculate $P(D \mid B)$

(b) **[4 points]** Calculate $P(B \mid D)$

Answer: Without any a-priori knowledge, let us assume that the first coin is equally likely to be either of the two coins. Let D denote the observation and B the event that the first coin is biased. Thus, $P(B) = P(\overline{B}) = \frac{1}{2}$,

$$P(D \mid B) = \left[\frac{2}{3}\right] \cdot \left[\binom{3}{2}\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)\right] = \frac{1}{4},$$

$$P(D \mid \overline{B}) = \left[\frac{1}{2}\right] \cdot \left[\binom{3}{2}\left(\frac{2}{3}\right)^2\left(\frac{1}{3}\right)\right] = \frac{2}{9}.$$

Using Bayes' theorem,

$$P(B \mid D) = \frac{P(D \mid B)P(B)}{P(D \mid B)P(B) + P(D \mid \overline{B})P(\overline{B})} = \frac{\frac{1}{4}}{\frac{1}{4} + \frac{2}{9}} = \frac{9}{17} > \frac{1}{2}.$$

Thus, the observation makes the first coin more likely to be biased.

Problem 10: Random Variable [10 points]

(a) **[6 points, 2 pts each]** Let $X_1, X_2, \ldots, X_{100}$ be the outcomes of 100 independent rolls of a fair die.

(i) **[2 points]** What are $\mathbb{E}(X_1)$ and $\text{var}(X_1)$?

(ii) **[2 points]** Define the random variable Y to be $X_1 - 2X_2 + X_3$. What are $\mathbb{E}(Y)$ and $\text{var}(Y)$?

(iii) **[2 points]** Define the random variable $Z = X_1 - X_2 + X_3 - X_4 + \cdots + X_{99} - X_{100}$. What are $\mathbb{E}(Z)$ and $\text{var}(Z)$?

(b) [**4 points**] Assume two scenarios.

Scenario 1: Roll 1 fair die and take the square of the number that comes up.

Scenario 2: Roll 2 fair dice and multiply the outputs.

Which expected value is higher? Explain your reason.

Answer: (a) (i) For a fair die, each outcome X_1 is uniformly distributed over $\{1, 2, 3, 4, 5, 6\}$. The expected value is:

$$\mathbb{E}(X_1) = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5$$

The variance is:

$$\text{var}(X_1) = \mathbb{E}(X_1^2) - [\mathbb{E}(X_1)]^2$$

where $\mathbb{E}(X_1^2) = \frac{1^2+2^2+3^2+4^2+5^2+6^2}{6} = \frac{91}{6}$, so:

$$\text{var}(X_1) = \frac{91}{6} - (3.5)^2 = \frac{91}{6} - \frac{12.25}{1} = \frac{91}{6} - \frac{73.5}{6} = \frac{17.5}{6} \approx 2.9167$$

(ii) For $X = X_1 - X_2$, since X_1 and X_2 are independent:

$$\mathbb{E}(X) = \mathbb{E}(X_1) - \mathbb{E}(X_2) = 3.5 - 3.5 = 0$$

$$\text{var}(X) = \text{var}(X_1) + \text{var}(X_2) = \frac{17.5}{6} + \frac{17.5}{6} = \frac{35}{6} \approx 5.8333$$

(iii) For $Y = X_1 - 2X_2 + X_3$, since X_1, X_2, X_3 are independent:

$$\mathbb{E}(Y) = \mathbb{E}(X_1) - 2\mathbb{E}(X_2) + \mathbb{E}(X_3) = 3.5 - 2 \cdot 3.5 + 3.5 = 0$$

$$\text{var}(Y) = \text{var}(X_1) + (-2)^2\text{var}(X_2) + \text{var}(X_3) = \frac{17.5}{6} + 4 \cdot \frac{17.5}{6} + \frac{17.5}{6} = \frac{105}{6} = 17.5$$

(iv) For $Z = X_1 - X_2 + X_3 - \cdots + X_{99} - X_{100}$, there are 50 positive terms and 50 negative terms:

$$\mathbb{E}(Z) = \sum_{i=1}^{50} \mathbb{E}(X_{2i-1}) - \sum_{i=1}^{50} \mathbb{E}(X_{2i}) = 50 \cdot 3.5 - 50 \cdot 3.5 = 0$$

$$\text{var}(Z) = \sum_{i=1}^{50} \text{var}(X_{2i-1}) + \sum_{i=1}^{50} \text{var}(X_{2i}) = 50 \cdot \frac{17.5}{6} + 50 \cdot \frac{17.5}{6} = 100 \cdot \frac{17.5}{6} = \frac{1750}{6}$$

(b) Let a and b be values of the two die rolls. The expected value of squaring the the first roll is $E(a^2)$, and the expected value of multiplying the two values is $E(ab)$. Since we can assume the two die rolls are iid, we have $E(ab) = E(a)E(b) = E(a)^2$

Since the difference between these two expected values is equal to the variance, $Var(a) = E(a^2) - E(a)^2$ it must be nonnegative. Therefore, we have $E(a^2) \geq E(a)^2$ so the expected value of squaring the result of one die roll is higher.

(c) Four ace divide the deck into five parts, each of which may contain from 0 to 48 cards. According the Symmetry Principle, all five have on average the same length which is, therefore, 48/5=9.6. The next card after an aceless run should be an ace - a card number 10.6.

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